REDLINE

Product Requirements Document: Evidence-backed race strategy calls

Feature name	REDLINE — HackSprint MVP
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1. PROBLEM

Race strategists decide under time pressure from several telemetry streams at once. The hard part is not reading a number; it is seeing that several small changes, none past its limit, add up to a developing problem. An AI strategy call that cannot show its evidence is hard to trust.

Problem statement. Race strategists need to know whether a strategy call is supported by the data, because risk often shows only when signals are read together over time, and a recommendation without visible evidence leads to late pit calls, lost time or a simulated failure.

What REDLINE is. A pit-wall copilot that reads synthetic telemetry, detects a developing risk, offers bounded options with stated assumptions, leaves the decision to the strategist, shows the simulated consequence, and scores the whole decision path. It never controls the car.

1a. Background & Dependencies

- **Event.** Daytona HackSprint #6, New York, October 16, 2026. Hacking 10:00 AM to 3:30 PM submission; finalists present in 3 minutes.
- **Data.** Synthetic telemetry and seeded race scenarios only. No live NASCAR, team or vehicle data.
- **Locked stack.** React + TypeScript + Vite front end with custom CSS; Python + FastAPI back end; REST between them; Gemini API (free tier) as the model. Details in the REDLINE Build Spec.

Dependency	Role in REDLINE	If unavailable
Daytona	Runs the baseline and REDLINE strategies on the same seed, in isolation	Local runner, same interface
Arize AX	Trace of inputs, call, decision and outcome; evaluators	Cached trace, rendered in-app
ElevenLabs	Race-engineer radio callout at CRITICAL, stating the call and the cited numbers	Pre-generated audio clips
SurrealDB	Lap history and past decisions	Local JSON
Gemini API (free tier)	Writes the plain-language explanation only	Template explanation from reason codes
AWS Bedrock (free event sandbox)	Optional swap for Gemini if enabled in the first 30 minutes	Stay on Gemini
Anchor Browser	Not used	—

No tool costs money. No sponsor tool is required by the rules; each is used because it does a real job and is judged for

a Best Use prize.

1b. Assumptions

- Privateer endurance teams and simulator programs, which lack factory-scale engineering, may value evidence-backed calls. This is a hypothesis; no interviews have been done.
- Telemetry, pit-window and AI race-engineer tools already exist (Bosch Motorsport, Crew Chief, Garage 61, ProRacer and others). REDLINE claims only the evidence, comparison and evaluation layer.
- Synthetic telemetry proves the software's behavior, not motorsport physics.
- A post-race debrief may be as valuable as the live call.
- **Falsification.** If strategists say existing tools already show enough evidence, or they will not share telemetry even locally, the niche must be reconsidered.

2. SUCCESS METRICS

Metric	Target
Prepared scenarios classified correctly (tire, fuel, engine)	3 of 3
Recommendations that cite only signals present and moving	100%
Recommendations that reach an outcome without a human decision	0
REDLINE detects the tire risk before the naive baseline	2 or more laps earlier
Missing or stale channel suppresses a confident call	100%
Ordered Arize trace with four evaluators passing	At least 1 live run
Primary demo, telemetry to recommendation	Under 90 seconds
Reset to the deterministic starting state	100%

3. REQUIREMENTS

P0 and P1 mark build order, not undecided scope. Everything listed ships if time allows; P0 ships first.

User Journey 1: Load and monitor a race

Load

- [P0] Strategist loads the tire-degradation scenario.
- [P1] Strategist switches to the fuel-window or engine-heat scenario.

Monitor

- [P0] Strategist sees lap, position, tire temperature, tire wear, fuel, engine temperature and lap delta, each with units.
- [P0] Each signal shows a short trend over recent laps.
- [P0] Strategist advances the race lap by lap.
- [P0] Units and freshness are validated before any analysis; a missing, stale or unknown-unit channel shows NO RECOMMENDATION and the reason.

User Journey 2: Detect developing risk

Classify

- [P0] Deterministic feature code compares the current lap with prior laps.
- [P0] Fixed rules classify risk as NORMAL, WATCH or CRITICAL.
- [P0] Conflicting signals cap the state at WATCH.

Show the evidence

- [P0] Strategist sees each triggering signal, its reason code and how it changed.
- [P0] Signals that did not change are listed as not cited.
- [P1] REDLINE estimates the remaining window in laps.

User Journey 3: Compare and recommend

Compare

- [P0] REDLINE shows CONTINUE, CONSERVE and PIT THIS LAP, each with projected trade-off, assumption and a supported flag.
- [P0] Only these three actions are valid; any other output is rejected.

Recommend

- [P0] A truth table picks the preferred option.
- [P0] Gemini writes a short explanation that may cite only signals in the evidence set; a violating explanation is replaced by the template.
- [P0] At CRITICAL, a race-engineer radio callout states the call and its numbers ("Box this lap. Rear tires up twelve degrees in three laps, wear accelerating.").
- [P0] Every callout has a pre-generated clip, so the demo never waits on the network.

User Journey 4: Decide and see the consequence

Decide

- [P0] Strategist chooses FOLLOW or IGNORE, with an optional rationale.
- [P0] No outcome is simulated before that choice.

Consequence

- [P0] The race advances under the chosen decision.
- [P0] FOLLOW, IGNORE and a naive single-threshold strategy are compared on the same seed.
- [P0] Every outcome is stamped SIMULATED.

User Journey 5: Debrief and evaluate

- [P0] Arize records the ordered trajectory: validate, features, risk, compare, recommend, human decision, outcome.
- [P0] The trace is shown inside REDLINE, not only in Arize.
- [P0] Four evaluators run: signal relevance, recommendation correctness, explanation faithfulness, trajectory order.
- [P1] One click opens the run in Arize.
- [P1] SurrealDB stores lap history and decisions.

User Journey 6: Reset

- [P0] One click restores the scenario to its seeded start.
- [P0] The demo runs repeatedly without editing any file.

4. APPENDIX

Design Decisions

Decision	Rationale
Risk and recommendation are deterministic; the model only explains	The call must be reproducible and testable against a truth table
Explanations are checked against the evidence set	Prevents the model from citing a signal that did not move
Synthetic telemetry with fixed seeds	No licensing or connectivity risk; the demo repeats exactly
Human FOLLOW / IGNORE is mandatory	Decision support only; the human keeps authority
A naive threshold baseline runs on the same seed	Shows what REDLINE adds, not just what it says
Three scenarios, one architecture	Shows generality without building a race platform
The trace is drawn in the app	The proof stays on screen if a network call fails
Positioned as an evaluation layer, not an AI race engineer	The broad category is crowded
ElevenLabs is P0 as a race-engineer radio call	Pit walls talk by radio; the voice carries the call and competes for the ElevenLabs prize
Gemini free tier is the model; Bedrock is an optional swap	Costs nothing and works tonight; the model only explains, so its job is small
Product label is REDLINE	Resolved
Visual identity: the Five Rings (sumi ink on washi, one vermilion)	Calm under pressure; distinct from SCRAM

Open Questions

- Proposed split: Aisling and Nicole lead REDLINE; Jason and Natalie lead SCRAM. Confirm tonight.
- Does the free AWS event sandbox include Bedrock? Test in the first 30 minutes; if not, stay on Gemini.
- Are the seeded thresholds in the Build Spec realistic enough for a strategist in the room? Adjust tonight only, never during the build.

Other Links

- REDLINE Build Spec (REDLINE_BUILD_SPEC.md) — binding for the coding agent
- SCRAM & REDLINE Storyboards canvas
- Daytona HackSprint #6 event brief